

## PHYS 515 - Python for Data Science and Scientific Computing

Semester: Spring 2026

End-Term Project

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### Project Overview

This end-term project is designed to assess your comprehensive understanding of machine learning and data analysis techniques covered throughout the semester. Each group has been assigned a unique dataset and is required to develop original problem statements, apply multiple methods learned in class, and produce a detailed technical report.

#### Key Objectives:

- Formulate well-defined, dataset-appropriate problem statements, just like exercise sheets
- Apply 3–4 different machine learning methods from the course curriculum
- Demonstrate understanding of Python packages through theoretical connections
- Present comprehensive analysis with visualizations and quantitative evaluation
- Communicate findings in a clear, professional technical report

### Allowed Machine Learning Methods

You are restricted to using **only** the following methods covered (or to be covered) in class:

#### Supervised Learning Methods:

1. **Linear Regression** – for continuous target prediction
2. **Logistic Regression** – for binary classification
3. **Softmax Regression (Multinomial Logistic Regression)** – for multi-class classification
4. **Polynomial Regression** – for non-linear relationships
5. **Regularization Techniques** – Ridge (L2) and Lasso (L1) only
6. **k-Nearest Neighbors (kNN)** – supervised classification/regression based on proximity

#### Unsupervised Learning Methods (Basic Applications Only):

1. **Principal Component Analysis (PCA)** – dimensionality reduction and feature extraction
2. **k-Means Clustering** – grouping similar data points

#### Important Notes:

- You must apply **at least 3–4 different methods** from the above list

- Unsupervised methods (PCA, k-Means) should be used primarily for preprocessing, or exploratory analysis, as only basics will be covered. If someone does not wish to use them, that's also perfectly fine and will not affect the grade.
- Advanced deep learning, ensemble methods (Random Forests, XGBoost), or methods not listed above are **not permitted**

## Python Package Requirements

**Allowed Libraries:** You may only use `scikit-learn` and `PyTorch` for implementing machine learning algorithms. Standard libraries for data manipulation (`NumPy`, `pandas`) and visualization (`matplotlib`, `seaborn`) are also permitted.

### Critical Requirement – Package Explanation:

For **every** `scikit-learn` or `PyTorch` function/class you use (e.g., `LogisticRegression`, `PCA`, `KNeighborsClassifier`, `StandardScaler`), you must include a brief explanation (2–3 sentences) in your report that connects the package to concepts covered in lectures.

### Example:

*“We use `sklearn.linear_model.Ridge` which implements  $L_2$  regularization by adding the penalty term  $\lambda \sum_{j=1}^p \omega_j^2$  to the ordinary least squares loss function. This prevents overfitting by shrinking coefficient magnitudes, as discussed in the lecture on regularization. The hyperparameter `alpha` in `sklearn` corresponds to our  $\lambda$  parameter.”*

This requirement ensures you understand the theoretical foundation rather than blindly applying tools or relying on AI-generated solutions.

## Report Structure & Requirements

There must be one single report for the project that each member of the group has to submit and each report must be **12–20 pages**. It should follow this structure:

### Part 1: Problem Statement Section (3–4 pages)

Following the format of course exercise sheets, clearly define multiple sub-problems that you will address. This section should include:

- Problem Motivation:** Why is this problem important? What real-world application does it serve?
- Research Questions:** Formulate 3–5 specific, well-defined questions your analysis will answer
- Sub-Problems:** Break down your research questions into concrete sub-problems (similar to Problem 1(a), 1(b), etc. in exercise sheets)
- Hypotheses:** State clear expectations about what you expect to find
- Success Criteria:** Define metrics to evaluate if your solutions are successful

### Quality Expectations:

- Problems should be original and formulated by the group tailored to their specific dataset
- Problems should require application of multiple course methods
- Problems should be challenging but feasible within project scope
- Avoid generic statements; be specific about what you will investigate

## **Part 2: Solution Section (8–16 pages)**

For each sub-problem defined in Part 1, provide comprehensive solutions including:

### **a) Data Exploration & Preprocessing:**

- Exploratory data analysis: distributions, correlations, patterns, outliers
- Missing value handling with justification
- Which features are going to be used and what is the selection logic
- Data visualization (scatter plots, histograms, heatmaps, etc.)
- Train-test-validation split strategy

### **b) Methodology:**

- Detailed explanation of each method applied
- Mathematical formulation (equations as covered in lectures)
- Python package explanations (2–3 sentences per package)
- Hyperparameter choices with justification
- Model training procedures

### **c) Results & Analysis:**

- Clear presentation of results (tables, plots, confusion matrices)
- Appropriate evaluation metrics (MSE,  $R^2$ , cross-entropy, regularization, etc.)
- Comparison between different methods
- Statistical significance or confidence intervals where applicable
- Critical interpretation: don't just report numbers, explain what they mean

### **d) Discussion:**

- What worked well? What didn't work? Why?
- How do different methods compare for your problem?
- Limitations of your approach
- Potential improvements or future directions

## **Report Quality Standards**

- All figures and tables must be clearly labeled with captions
- Use proper mathematical notation consistent with lectures
- Maintain logical flow and coherent organization
- Reference/cite any external sources (dataset documentation, books, papers, etc.)

## Grading Rubric (100 Points Total)

Your project will be evaluated according to the following criteria:

| Component                                      | Points    | Evaluation Criteria                                                                           |
|------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------|
| <b>1. Problem Formulation &amp; Motivation</b> | <b>30</b> |                                                                                               |
| Clear problem statement(s)                     | 10        | Well-defined, specific to dataset, avoiding generic descriptions                              |
| Research questions & hypotheses                | 6         | Thoughtful, feasible questions requiring multiple methods                                     |
| Real-world relevance                           | 4         | Strong justification for problem importance and applications                                  |
| Sub-problem structure                          | 5         | Logical breakdown into solvable components following exercise sheet format                    |
| Success criteria definition                    | 5         | Clear, measurable metrics for evaluation                                                      |
| <b>2. Data Exploration &amp; Preprocessing</b> | <b>15</b> |                                                                                               |
| Exploratory data analysis                      | 5         | Comprehensive analysis of distributions, correlations, patterns                               |
| Data cleaning & handling                       | 4         | Appropriate treatment of missing values, outliers with justification                          |
| Feature selection                              | 3         | Creative, well-justified feature transformations or selections                                |
| Visualization quality                          | 3         | Clear, informative plots properly labeled and integrated                                      |
| <b>3. Methodology &amp; Implementation</b>     | <b>25</b> |                                                                                               |
| Method selection & application                 | 10        | Appropriate use of <i>at least</i> 3–4 course methods; correct implementation                 |
| Mathematical explanation                       | 5         | Proper equations, notation consistent with lectures                                           |
| Python package explanations                    | 5         | Every scikit-learn/PyTorch function explained with 2–3 sentences connecting to lecture theory |
| Hyperparameter justification                   | 4         | Well-reasoned choices; evidence of experimentation for choosing hyperparameters               |
| Multiple approach comparison                   | 3         | Testing different methods/parameters with comparative analysis                                |
| <b>4. Results &amp; Analysis</b>               | <b>20</b> |                                                                                               |
| Results presentation                           | 5         | Clear tables, plots, confusion matrices appropriately formatted                               |
| Evaluation metrics                             | 5         | Appropriate metrics correctly calculated for all methods                                      |
| Comparative analysis                           | 5         | Meaningful comparison between methods with insights                                           |

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| Component                                    | Points    | Evaluation Criteria                                               |
|----------------------------------------------|-----------|-------------------------------------------------------------------|
| Critical interpretation                      | 5         | Deep analysis beyond reporting numbers; addresses "why" questions |
| <b>5. Report Quality &amp; Communication</b> | <b>10</b> |                                                                   |
| Writing clarity & organization               | 4         | Logical flow and clear explanations                               |
| Figure/table labeling                        | 2         | All visuals properly captioned and referenced in text             |
| Professional presentation                    | 2         | Clean formatting, adherence to guidelines                         |
| Length & structure compliance                | 2         | Meets 12–20 page requirement with proper section divisions        |

### Grading Notes:

- **Package Explanations (8 points):** This is a critical component. Missing or superficial explanations will result in significant point deductions. The goal is to demonstrate understanding, not just tool usage.
- **Partial Credit:** Within each component, partial credit will be awarded based on completeness and quality.
- **Problem Formulation Weight:** Note that problem formulation carries the highest individual weight (30 points). Creative, well-motivated problems that effectively use the dataset will be rewarded.
- **Method Diversity:** Using only 2 methods or failing to apply them meaningfully will result in reduced methodology scores.
- **Plagiarism & AI Detection:** The package explanation requirement serves as an AI-detection mechanism. Unexplained code or inconsistencies between explanations and implementation will lead to point deductions.

## Submission Guidelines

### Report Format:

- The entire report must be typed in  $\text{\LaTeX}$  and submitted as a PDF.
- Use a professional document class (e.g., `article`, `scrartcl`) with readable font size (11pt or 12pt).
- Plots should be embedded directly in the  $\text{\LaTeX}$  document using `includegraphics`.
- Code snippets may be included using the `listings` or `minted` package for syntax highlighting. However, this is not essential because full code files must be submitted separately (see below).

**Submission Requirements:**

Submit a single compressed folder (e.g., .zip file) containing:

- Your main PDF report (compiled from L<sup>A</sup>T<sub>E</sub>X), including all figures
- All Python code files (.py or .ipynb)
- **Do not submit** L<sup>A</sup>T<sub>E</sub>X source files (.tex)

— **Due Date:** Wednesday, May 10, 2026 by midnight

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**Good luck with your projects! This is your opportunity to showcase everything you've learned this semester.**